

SAMRAT MALISETTI

Software Engineer — Agentic AI & LLM Systems

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SUMMARY

Software engineer with **5+ years of experience**, currently building production AI agent systems at ADP that serve **41M+ employees across 1.1M clients**. Specialty: making LLM agents safe to trust with real actions — guardrails, human-in-the-loop flows, eval frameworks, durable memory, and observability at every layer. The more important the answer, the less the AI should improvise: models interpret the human, deterministic code does the math, evals gate every release.

EXPERIENCE

Software Engineer — ADP

May 2025 – Present

Pasadena, CA · Production AI agents on ADP Assist (41M+ employees, 1.1M clients)

- Core engineer on the **WFM Clocking Assistant** — a conversational agent that lets employees clock in/out, manage breaks, handle attestations, and complete department transfers through natural language, with compliance checks built in.
- Architected the agent on the **Strands Agent SDK** (agent loop, tool system, steering hooks), routed through ADP's AI Gateway via Azure OpenAI, with an intent orchestrator deciding where each query goes.
- Designed **human-in-the-loop confirmation flows** with hash validation, expiry, and audit logging, plus steering guardrails enforcing business rules — an agent that punches a timecard wrong is a payroll problem, not a chatbot quirk.
- Implemented durable cross-conversation memory with **AgentCore Memory on AWS Bedrock**, persisting only what's needed.
- Cut time-to-first-token cost by **75%** with a 2-layer prefix-caching strategy (global static + per-session context).
- Built an automated **eval framework** that tests agent behavior before anything ships — intent mapping, action parameters, HITL confirmations, steering blocks, and multi-turn conversations.
- Added production observability with **Langfuse**: per-request tracing of tool calls, latency, token usage, and cost.
- Reduced CI/CD deployment time from **80 minutes to 12** on Concourse + Kubernetes.

Full Stack Engineer (Contract) — GorgeousTV

Jun 2024 – Feb 2025

Los Angeles, CA · Remote

- Analyzed cloud resource usage across GCP and AWS to identify cost savings and performance improvements as part of an Azure migration initiative; evaluated multi-cloud compatibility with **Azure Migrate**.
- Configured Azure Virtual Networks, subnets, and network security groups, mapping existing GCP infrastructure to equivalent Azure architecture for seamless migration.
- Deployed containerized services with **Kubernetes + Docker**, implementing scalable deployment strategies to improve reliability and uptime.
- Automated infrastructure monitoring and logging with **Prometheus + Grafana**, improving observability and incident response.
- Managed IAM roles and access policies across GCP, AWS, and Azure, enforcing consistent access control throughout the migration.

Software Engineer — ZeMoSo Technologies

May 2021 – Aug 2022

Hyderabad, India · Field Squared field-service platform (React Native)

- Designed and implemented an **offline request queue** so field technicians could keep working without network access, syncing data automatically once connectivity was restored.
- Enabled enterprises to automate field-service operations with features for task scheduling, data collection, and workflow management; improved UI/UX from direct technician feedback.
- Optimized GraphQL middleware, cutting average latency and response times by **25%**.

Software Engineer — KeeStep

Jan 2019 – Feb 2021

Hyderabad, India · College management platform, built end-to-end over 2+ years

- Developed a comprehensive college management application with React, NestJS, Express, and MongoDB to streamline academic workflows and data management across multiple user roles.
- Architected **RESTful APIs and database schemas from scratch**, building a robust and scalable backend architecture.
- Designed modular, reusable UI components in React, optimizing the application for scalability, maintainability, and user experience.
- Implemented **role-based access control on every API** using custom middlewares and decorators; integrated authentication and authorization with PassportJS for secure multi-role access.
- Built real-time security protocols with NestJS Guards and **WebSockets**, improving responsiveness in dynamic environments.
- Optimized performance with lazy loading, task scheduling, and cookie management.

SELECTED PROJECTS

Agent Inc. — RL Environment That Teaches a Model to Run a Business

YC × HUD RL Hackathon · 2026

An RL environment that trains a model to run a business end-to-end — read the brief, research, make a truthful priced offer, ship a deliverable — scored by a hybrid reward (**70% deterministic + 30% LLM judge**) that only pays for honest, complete work. On-policy GRPO lifted an open model (Qwen3.5-4B) **0.327** → **0.647** — nearly doubling its score and closing ~half the gap to a frontier model, on 24 held-out scenarios it never trained on. Python · HUD · FastMCP · GRPO.

Amparo AI — Multilingual Insurance Voice Agent

YC Hackathon · 2026

Phone-call agent that answers health-plan questions in any language: **24 parallel retrievals at <100ms** per turn → cited 4-plan cost comparison in under 2 seconds. The LLM never computes a number — deterministic code does the math, every fact is cited to its exact location in the real plan PDF, and a regex-based clinical-handoff guardrail routes medical questions to a doctor. Exposed a "cheapest" plan actually costing **\$22k more per year** due to out-of-network rules; 85 tests. LiveKit · Python · SIP · TTS/STT.

Engram — Shared Account Brain for Sales Teams

Agentic AI Hackathon SF · 2026

iMessage agent that detects contradictions against a team fact ledger, **supersedes stale facts with full version lineage** (v1→v6), and delivers source-attributed account briefs on demand. Express/TypeScript spine with extract → reconcile → answer pipelines and a live memory panel.

Specforge — Speak an API Into Existence

Google I/O Hackathon · 2026

A Managed Gemini Agent writes a backend from plain English, **tests it in its own sandboxed VM until green**, and a FastAPI control plane hot-reloads it behind a public URL; the service evolves across conversational turns ("now add auth").

SKILLS

AGENTIC AI	Strands Agent SDK, AWS Bedrock · AgentCore Memory, Azure OpenAI, Claude API, Gemini, multi-agent orchestration, RAG & hybrid retrieval, eval frameworks, LLM observability (Langfuse), HITL guardrails, prompt caching, MCP, voice agents (LiveKit · STT/TTS)
LANGUAGES	TypeScript, JavaScript, Python, Java, SQL
WEB & MOBILE	React, Next.js, React Native, Expo, Node.js, NestJS, Express, FastAPI, GraphQL, WebRTC, PostgreSQL, MongoDB
CLOUD & DEVOPS	AWS, GCP, Azure, Kubernetes, Docker, CI/CD (Concourse), Prometheus, Grafana

EDUCATION

M.S. Computer Science — California State University, Long Beach

2022 – 2024

- Open-source contributions during graduate study: SuperMemory, Polar, TensorZero — UI/UX, API integration, performance.